

BANFF, AB, Canada

WMO: 711220

Lat: 51.1936N

Lon: 115.5525W

Elev: 1397

StdP: 85.63

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-29.7	-25.9	-34.0	0.2	-29.5	-30.1	0.3	-25.5	7.2	-6.9	6.6	-6.3	1.1	220	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	27.7	13.5	25.8	13.0	23.7	12.4	14.8	23.7	13.9	22.7	13.1	21.6	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	10.4	16.3	10.8	9.6	15.3	9.8	8.9	14.7	46.2	23.6	43.4	22.8	41.0	21.8	18.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.5	5.1	DB	-36.1	30.4	4.4	1.9	-39.2	31.8	-41.8	32.9	-44.2	34.0	-47.4	35.4	(4)
(5)			WB	-35.8	16.6	3.8	1.1	-38.6	17.3	-40.8	18.0	-43.0	18.6	-45.8	19.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-8.3	-6.8	-2.3	2.5	7.2	11.0	14.7	13.9	9.4	3.4	-3.6	-9.1	(6)
(7)		DBStd	9.81	7.75	6.97	6.30	4.12	3.79	3.08	3.12	3.17	3.99	4.36	6.20	7.14	(7)
(8)		HDD10.0	3048	567	471	382	224	104	23	3	6	59	207	409	592	(8)
(9)		HDD18.3	5707	826	704	640	473	346	219	117	140	269	463	659	850	(9)
(10)		CDD10.0	389	0	0	0	1	16	54	151	126	40	2	0	0	(10)
(11)		CDD18.3	8	0	0	0	0	0	1	6	2	0	0	0	0	(11)
(12)		CDH23.3	494	0	0	0	0	4	34	235	185	36	0	0	0	(12)
(13)		CDH26.7	94	0	0	0	0	0	6	53	33	3	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.8	2.5	2.4	2.3	2.1	2.0	2.1	1.9	2.0	2.4	2.7	2.8	(14)
(15)	Precipitation	PrecAvg	487	33	24	23	37	60	62	52	54	48	30	31	36	(15)
(16)		PrecMax	617	139	109	50	73	152	106	114	149	84	88	61	99	(16)
(17)		PrecMin	352	0	4	7	9	19	18	11	8	7	7	3	1	(17)
(18)		PrecStd	74	27	23	13	15	33	23	29	29	21	18	18	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	7.8	13.2	19.1	23.8	27.0	30.4	29.6	26.8	18.8	10.7	5.7	(19)
(20)			MCWB	2.3	1.9	4.3	8.1	10.9	13.1	14.5	14.0	12.5	9.2	5.2	2.3	(20)
(21)		2%	DB	4.1	5.3	9.5	15.7	21.3	23.7	28.3	27.3	24.1	15.2	6.9	2.8	(21)
(22)			MCWB	0.9	0.9	3.0	6.5	9.7	12.0	13.8	13.3	11.9	7.7	3.3	0.1	(22)
(23)		5%	DB	2.4	3.4	7.4	12.6	18.8	21.2	26.1	25.5	21.4	12.6	5.0	1.1	(23)
(24)			MCWB	-0.3	-0.3	1.9	4.9	8.8	10.9	13.3	12.9	11.0	6.4	2.0	-1.0	(24)
(25)		10%	DB	0.7	1.8	5.3	10.0	16.0	18.9	23.8	23.2	18.1	10.2	3.3	-0.6	(25)
(26)			MCWB	-1.5	-1.4	0.9	3.6	7.7	10.2	12.9	12.3	9.9	5.1	1.0	-2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.9	5.1	8.4	11.9	14.5	16.6	15.7	13.7	10.1	5.6	2.5	(27)
(28)			MCDWB	5.5	6.2	11.1	17.5	20.8	23.1	25.2	24.3	23.1	17.4	9.3	4.8	(28)
(29)		2%	WB	1.2	1.4	3.7	7.0	10.5	13.0	15.2	14.6	12.5	8.4	4.0	0.5	(29)
(30)			MCDWB	3.6	4.6	8.3	14.7	18.7	21.1	24.0	23.6	21.9	14.0	6.2	2.3	(30)
(31)		5%	WB	-0.2	0.0	2.5	5.5	9.4	11.8	14.2	13.7	11.4	6.9	2.5	-0.8	(31)
(32)			MCDWB	2.0	3.1	6.6	11.5	17.2	19.3	22.9	22.7	20.1	11.7	4.6	1.0	(32)
(33)		10%	WB	-1.3	-1.2	1.2	4.0	8.2	10.8	13.4	12.9	10.2	5.4	1.2	-2.3	(33)
(34)			MCDWB	0.7	1.6	4.9	9.3	15.3	17.4	21.9	21.4	17.6	9.6	3.1	-0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	11.2	10.7	11.4	12.7	12.4	14.7	15.2	13.4	10.4	8.0	8.0	(35)
(36)			MCDBR	7.8	10.5	12.3	16.5	18.7	18.1	20.1	20.6	20.5	14.7	8.3	6.7	(36)
(37)			MCWBR	5.3	6.9	7.0	8.8	9.1	8.3	8.2	8.5	9.6	8.0	5.4	5.0	(37)
(38)		5% WB	MCDBR	7.1	9.9	11.4	14.7	16.6	15.4	16.9	17.7	18.6	13.8	7.7	6.6	(38)
(39)			MCWBR	5.2	6.7	6.8	8.2	8.3	7.4	7.5	7.9	9.0	7.8	5.3	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.246	0.258	0.285	0.337	0.351	0.369	0.357	0.370	0.328	0.286	0.254	0.236	(40)	
(41)		taud	2.370	2.360	2.299	2.168	2.289	2.325	2.375	2.351	2.449	2.549	2.460	2.362	(41)	
(42)		Ebn at Noon	804	890	920	899	902	885	890	859	864	850	792	758	(42)	
(43)		Edh at Noon	59	79	103	133	124	121	113	110	89	66	54	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.10	2.05	3.26	4.53	5.30	5.57	6.23	5.18	3.64	2.26	1.19	0.87	(44)	
(45)		RadStd	0.07	0.14	0.22	0.33	0.45	0.46	0.44	0.37	0.42	0.22	0.08	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page